IPS-YOLO: Iterative Pseudo-Fully Supervised Training of YOLO for Weakly Supervised Object Detection in Remote Sensing Images

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***Abstract*— Weakly supervised object detection (WSOD) in remote sensing images (RSIs) only requires image-level labels, greatly reducing the cost of manual annotations. Recently, the pseudo-fully supervised object detection (pseudo-FSOD) models give better performance than traditional WSOD models based on multiple instance learning. However, the existing pseudo-FSOD models still have two problems that need to be addressed. First, existing models tend to focus on the salient parts of object rather than the whole object. Second, the pseudo ground truth (PGT) cannot be continuously improved during the training process. For the first problem, a filtering and weighted synthesis guided by category confidence score (FWSC) strategy is proposed to generate PGT. The FWSC strategy first removes the instances with low category confidence score (CCS), then, to cover the whole object as much as possible, any two instances are weighted and synthesized according to their CCSs if they have very high spatial overlap, otherwise, the nonmaximum suppression (NMS) operation is conducted. For the second problem, an iterative refinement (IR) scheme of PGT is proposed. Specifically, the PGT instances produced by the FWSC strategy are first used to train a YOLO model, then a filtering and weighted synthesis guided by IoU (FWSI) strategy employs the detection results inferred from the trained YOLO model to refine the PGT instances, and the above two steps can be repeated multiple times. Furthermore, three improvement strategies are proposed to enhance the traditional baseline WSOD model in proposals generation, the selection of augmented samples, and the definition of pseudo labels, respectively. The ablation studies demonstrate the effectiveness of FWSC strategy, IR scheme of PGT, and the three improvement strategies of baseline model. The comparisons with popular WSOD models show that our model gives the best results on the NWPU VHR-10.v2 and DIOR datasets. The source codes have been released at https://github.com/qxlzengli/IPS-YOLO**

***Index Terms*— Filtering and weighted synthesis guided by category confidence score (FWSC), filtering and weighted synthesis guided by IoU (FWSI), iterative refinement (IR) of pseudo ground truth (PGT), remote sensing image (RSI), weakly supervised object detection (WSOD).**

I. Introduction

OBJECT detection in remote sensing images (RSIs) [1], [2], [3], [4], [5], [6], [7], [8], [9], [10] has been applied to numerous practical situations, such as danger prediction [11], [12], [13], land resource survey [14], [15], [16], and urban construction [17], [18]. Fully supervised object detection (FSOD) in RSIs [19], [20], [21] has achieved high accuracy by utilizing manually annotated instance-level labels. However, the instance-level manual annotations of numerous RSIs require significant human resources and time cost [22], [23]. Therefore, weakly supervised object detection (WSOD) in RSIs [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], which only requires image-level labels, has been introduced to reduce the cost of manual annotations.

The existing WSOD models can be divided into two categories, i.e., the traditional multiple instance learning (MIL)-based models, and the recently emerged pseudo-fully supervised object detection (pseudo-FSOD) models. The milestone traditional WSOD model, i.e., OICR [34], utilizes selective search (SS) to generate object proposals, and then employs MIL to produce pseudo label for each proposal, which is used to train instance classifier refinement (ICR) branches. The pseudo-FSOD models use the results generated by WSOD models as pseudo ground truth (PGT) to guide the training of FSOD models. Due to the advantages of FSOD models in terms of detection accuracy, inference speed, and the speed of technological iteration, the pseudo-FSOD models have demonstrated excellent performance and thus attracted more attentions; however, there are still two issues that need to be addressed.

The first issue is that pseudo-FSOD models tend to focus on the most salient parts rather than the whole object. The PGT for training pseudo-FSOD is typically generated by using the following methods [35], [36]. First, the instances output from WSOD model whose category confidence scores (CCSs) are lower than predefined threshold are eliminated. Then, a nonmaximum suppression (NMS) operation is imposed on the remaining instances to remove the redundant instances. The

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Fig. 1. Illustration of motivations of (a) FWSC and (b) IR strategies.

TABLE I
ABLATION STUDY OF IMPROVED BASELINE MODEL ON THE NWPU VHR-10.V2 AND DIOR DATASETS

Baseline	SAMPG	EPS	SL	NWPU VHR-10.v2 mAP (%)	DIOR mAP (%)
✓				51.52	22.18
✓	✓			59.68	25.82
✓		✓		60.30	26.28
✓			✓	61.89	26.50
✓	✓	✓	✓	65.19	27.20

TABLE II
ABLATION STUDY OF FWSC AND IR ON THE NWPU VHR-10.V2 AND DIOR DATASETS

Improved Baseline	FWSC	IR	NWPU VHR-10.v2 mAP (%)	DIOR mAP (%)
✓			65.19	27.20
✓	✓		72.30	29.38
✓		✓	73.60	29.50
✓	✓	✓	76.20	30.40

above scheme ensures that the generated PGT instances have high CCS. However, the detection results with high CCS given by current WSOD models usually focus on the most salient parts of object. Consequently, pseudo-FSOD models trained by aforementioned PGT instances also tend to focus on the most salient parts of object. This problem is more serious in bird's-eye viewing RSIs which have less texture information, because the salient parts of object have a higher proportion of key structure characteristics. As shown in Fig. 1(a), the baseline model focuses on the salient parts of baseball court.

The second issue is that the quality of PGT cannot be continuously improved. So far, only the noisy PGT instances are removed and the rest of PGT instances are unchanged during the training process. For example, SoS [35] removes PGT instances whose training loss values exceed predefined threshold, and uses the remaining PGT instances as a clean training set to train the FSOD model. However, the pseudo-FSOD model trained by current PGT may lead to the missed detection of objects, especially when images contain multiple instances with same category. The reasons are as follows. As discussed in the last paragraph, the current PGT is generated from the traditional WSOD models, and the WSOD models driven by image-level labels tend to focus on the most salient instance of each category because the image-level label of each category can be determined by the presence of only one instance of that category. Compared with natural scene images, the missed detection problem is more serious in RSIs, because RSIs have a long imaging distance which usually leads to the dense distribution of multiple objects with the same category. As shown in Fig. 1(b), the baseline model omits some airplanes.

To address the above issues and accelerate the inference speed, an iterative pseudo-fully supervised training scheme of YOLO (IPS-YOLO) [37] model is proposed for WSOD in RSIs. The major contents of IPS-YOLO are as follows.

To address the first issue, IPS-YOLO proposes a filtering and weighted synthesis guided by CCS (FWSC) strategy for generating PGT. The primary distinction between the FWSC strategy and current strategies lies in that any two instances with high CCS and spatial overlap are weighted and synthesized according to their CCSs instead of eliminating the one with lower CCS, which ensures that the generated PGT instances can cover the entire object as much as possible. As a matter of fact, there is a high probability that two instances with same category and high spatial overlap cover the same object. As shown in Fig. 1(b), two small instances with high spatial overlap are fused by using the FWSC strategy to produce a better instance which covers the larger area of baseball court.

To handle the second issue, IPS-YOLO introduces an iterative refinement (IR) scheme of PGT. This scheme first uses the PGT to train a YOLO model, and then a filtering and weighted synthesis guided by IoU (FWSI) strategy is proposed, which uses the detection results given by the trained YOLO model to update the PGT. The above process can be repeated many times; consequently, the quality of PGT is continuously improved and the capability of the YOLO model is improved accordingly. As a matter of fact, although the initial PGT tends to focus on the most salient instance of each category, the YOLO model trained by PGT can learn the key features of each category, and the detection results of the YOLO model may discover more instances of each category; therefore, using FWSI strategy to iteratively refine PGT can effectively alleviate the missed detection problem. As shown in Fig. 1(b), all airplanes are detected by using our IR scheme.

In addition, the traditional baseline WSOD model is also improved in this article. The improvement scheme mainly includes three aspects. First, the segment anything model (SAM) [38] is used to enhance the quality of object proposals. Second, the pseudo labels of some instances are redefined by soft labels (SLs) rather than one-hot labels in ICR branches. Finally, the training samples are augmented (rotation, flipping, and scaling), and each type of augmented samples is randomly selected with equal probability, which ensures that the three kinds of augmented samples impose same influence on the training procedure to enhance the robustness of IPS-YOLO.

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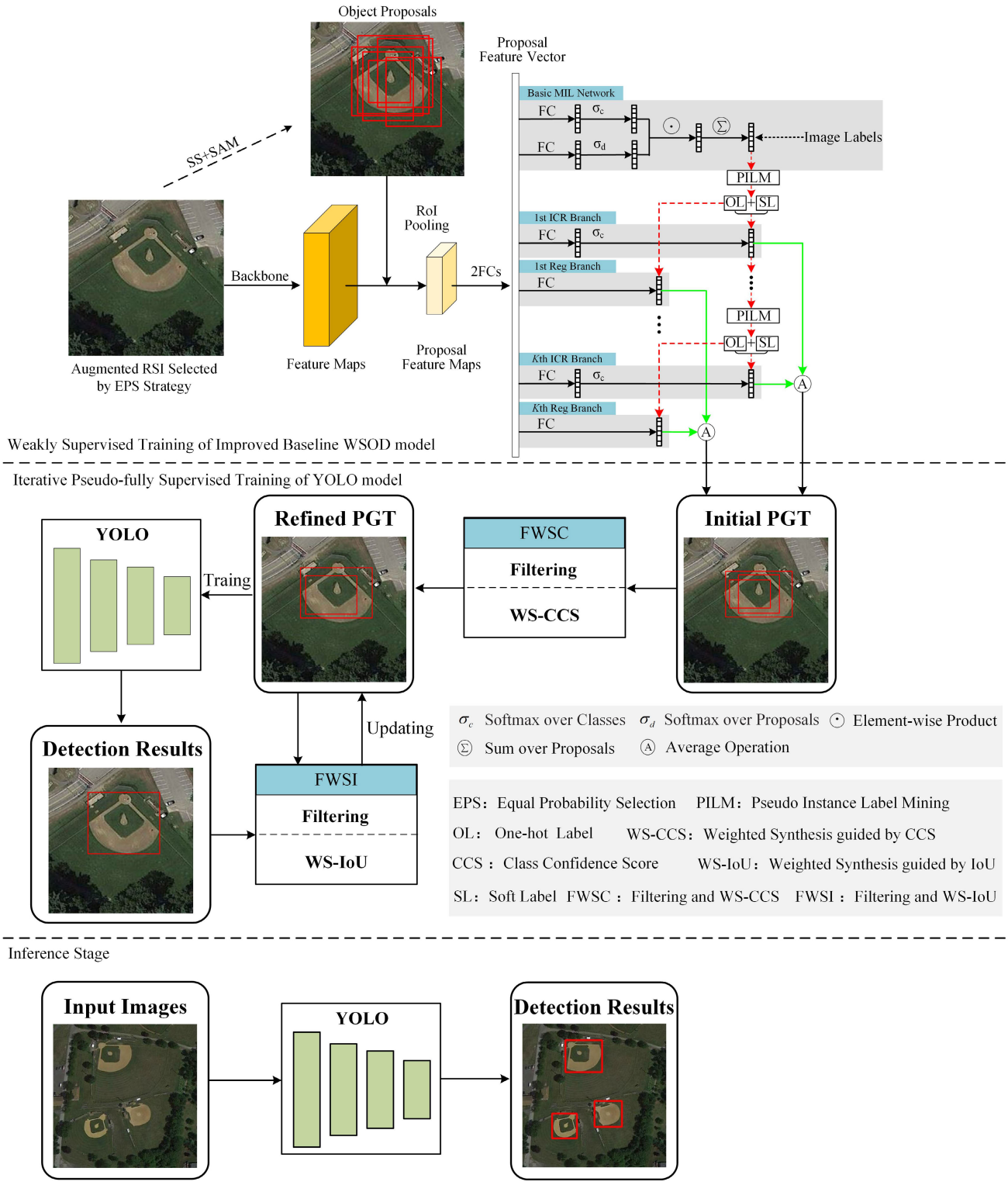


Fig. 2. Framework of our IPS-YOLO model.

The main contributions of this article are summarized as follows.

1) To address the issue that the current pseudo-FSOD models tend to focus on the most salient parts of object, an FWSC strategy is proposed to generate PGT, where any instances with high CCS and spatial overlap are weighted and synthesized according to their CCSs.
2) To address the issue that the generated PGT cannot be continuously improved during the training of pseudo-FSOD models, an IR scheme of PGT is introduced, which uses the proposed FWSI strategy and detection results given by the trained YOLO model to iteratively refine the PGT.
3) The object proposals, the definition of pseudo labels, and the selection scheme of augmented samples are

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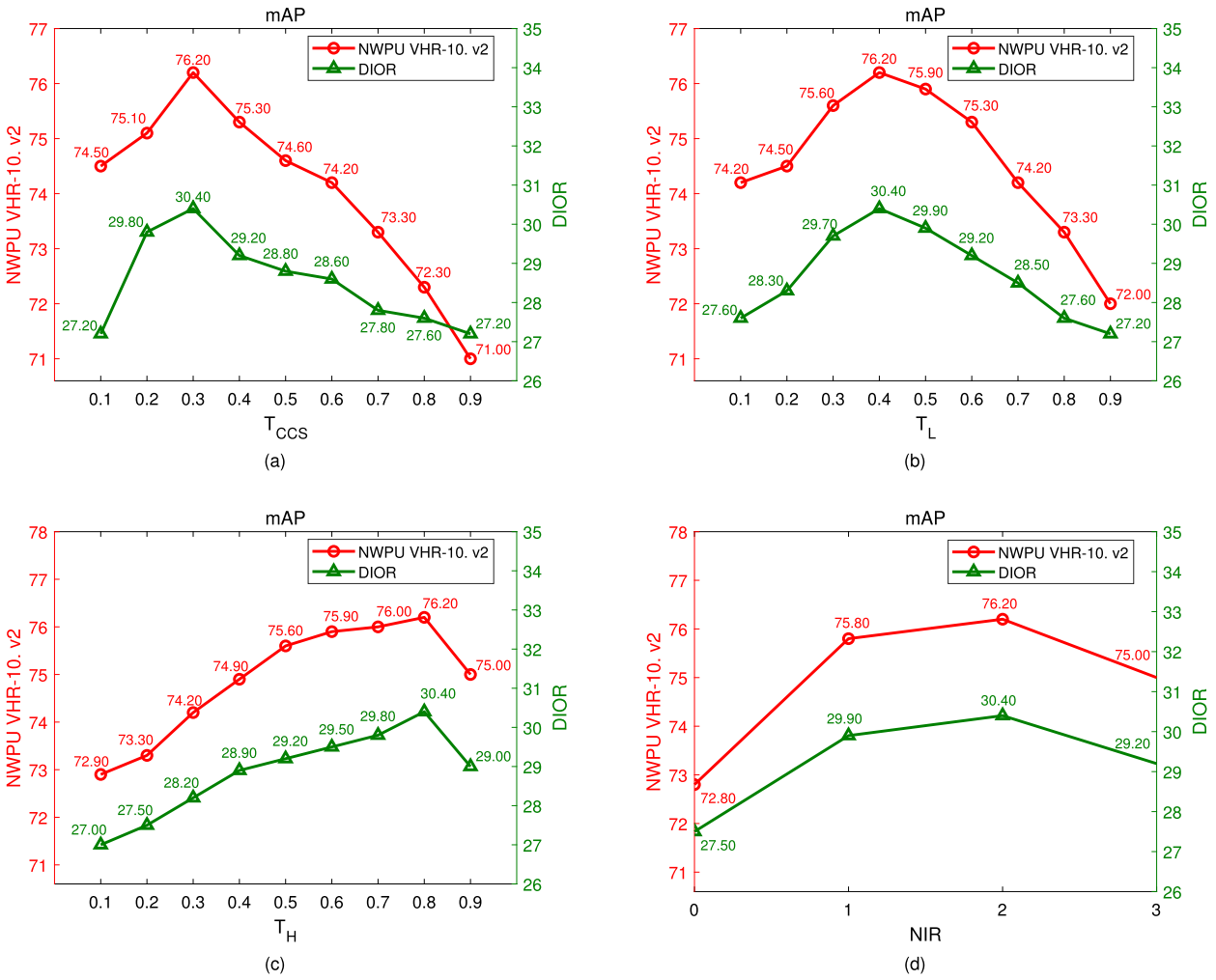


Fig. 3. Parameter analysis of (a) T_{CCS}, (b) T_L, (c) T_H, and (d) NIR in terms of mAP on the NWPU VHR-10.v2 and DIOR dataset.

improved to enhance the capability of traditional baseline WSOD model.

4) The IPS-YOLO model achieves state-of-the-art (SOTA) performance on the NWPU VHR-10.v2 and DIOR datasets.

II. Related Work

This section introduces the baseline model used in this article as well as existing WSOD models.

A. Baseline WSOD Model

Early WSOD models typically use OICR [34] as the baseline. Later, it is discovered that adding a regression branch to each ICR branch could significantly improve the performance of WSOD. As a result, the above improved OICR becomes the commonly used baseline model, and our IPS-YOLO model follows the same way.

1) Weakly Supervised Deep Detection Network: Weakly supervised deep detection network (WSDDN) is shown in the upper part of Fig. 2. First, the SS is used to generate N object proposals, i.e., $R = \{r_1, \ldots r_n, \ldots r_N\}, n \in [1, N]$, then a feature vector of each proposal is obtained by successively passing through region of interest (ROI) pooling operation and two fully connected (FC) layers. Subsequently, the feature vector of each proposal is fed into a classification (detection) branch to obtain a classification (detection) score matrix, where the classification (detection) branch includes an FC layer and a softmax over class (proposal) operation, and the ultimate score matrix of each proposal is derived from the elementwise multiplication of classification and detection matrixes. Afterward, the image-level CCS of input RSI in the cth category, denoted as IS_c, is calculated through following equation:

$$\mathrm{IS_c} = \sum_{n=1}^{N} X_{c,n}, c \in [1, C] \tag{1}$$

where C refers to the number of object categories, and $x_{c,n}$ represents the element located at the cth row and nth column of aforementioned score matrix. Finally, the loss function of WSDDN, denoted as L_W, can be formulated as follows:

$$L_W = -\sum_{c=1}^{C} (y_c \log \mathrm{IS}_c + (1 - y_c) \log(1 - \mathrm{IS}_c)) \tag{2}$$

where $y_c = 1$ or 0 indicates whether or not the input RSI contains the cth category.

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TABLE III
COMPARISON WITH POPULAR METHODS ON THE NWPU VHR-10.V2 DATASET IN TERMS OF MAP (%)

Method	Airplane	Ship	Storage tank	Baseball Diamond	Tennis Court	Basketball Court	Ground Track field	Harbor	Bridge	Vehicle	mAP
WSDDN [39]	30.08	41.72	34.98	88.90	12.86	23.85	99.43	13.94	1.92	3.60	35.12
OICR [34]	13.66	67.35	57.16	55.16	13.64	39.66	92.80	0.23	1.84	3.73	34.52
PCL [42]	26.00	63.76	2.50	89.80	64.45	76.07	77.94	0.00	1.30	15.67	39.41
MIST [41]	69.68	49.16	48.55	80.91	27.08	79.85	91.34	46.99	8.29	13.36	51.52
DCL [45]	72.70	74.25	37.05	82.64	36.88	42.27	83.95	39.57	16.82	35.00	52.11
PCIR [56]	90.78	78.81	36.40	90.80	22.64	52.16	88.51	42.36	11.74	35.49	54.97
MIG [57]	88.69	71.61	75.17	94.19	37.45	47.68	100.00	27.27	8.33	9.06	55.95
TCA [58]	89.43	78.18	78.42	90.80	35.27	50.36	90.91	42.44	4.11	28.30	58.82
SAE [32]	82.91	74.47	50.20	96.74	55.66	72.94	100.00	36.46	6.33	31.89	60.72
SPG [31]	90.42	81.00	59.53	92.31	35.64	51.44	99.92	58.71	16.99	42.99	62.80
MHQ-PSL [24]	87.60	81.00	57.30	94.00	36.40	80.40	100.00	56.90	9.80	35.60	63.80
SGPLM-IR [25]	90.70	79.90	69.30	97.50	41.60	77.50	100.00	44.40	17.20	33.50	65.20
PIIFR [59]	90.80	81.60	56.60	91.70	51.90	69.50	100.00	53.40	16.30	40.50	65.20
HIMWR [47]	90.81	80.52	73.42	96.59	47.35	78.94	100.00	43.89	18.35	33.51	66.24
RINet [60]	90.30	86.30	79.60	90.70	58.20	80.40	100.00	57.70	18.90	41.60	70.40
AE-IS [30]	91.00	88.20	78.30	93.20	60.60	82.40	100.00	60.40	19.60	45.80	72.00
IPS-YOLOm (Ours)	99.45	87.88	49.52	97.99	87.85	92.63	80.19	89.38	1.01	76.42	**76.20**

TABLE IV
COMPARISON WITH POPULAR METHODS ON THE NWPU VHR-10.V2 DATASET IN TERMS OF CORLOC (%)

Method	Airplane	Ship	Storage tank	Baseball Diamond	Tennis Court	Basketball Court	Ground Track field	Harbor	Bridge	Vehicle	CorLoc
WSDDN [39]	22.32	36.81	39.95	92.48	17.96	24.24	99.26	14.83	1.69	2.89	35.24
OICR [34]	29.41	83.33	20.51	81.76	40.85	32.08	86.60	7.41	3.70	14.44	40.01
PCL [42]	11.76	50.00	12.82	98.65	84.51	77.36	90.72	0.00	9.26	15.56	45.06
MIST [41]	90.20	82.50	80.30	98.60	48.50	87.40	98.30	66.50	14.60	35.80	70.30
PCIR [56]	100.00	93.06	64.10	99.32	64.79	79.25	89.69	62.96	13.26	52.22	71.87
MIG [57]	97.79	90.26	87.18	98.65	54.93	64.15	100.00	74.07	12.96	21.57	70.16
TCA [58]	96.91	91.78	95.13	88.65	66.90	62.88	95.98	54.18	19.63	55.50	72.76
SAE [32]	97.06	91.67	87.81	98.65	40.86	81.13	100.00	70.37	14.81	52.22	73.46
SPG [31]	98.06	92.67	70.08	99.65	51.86	80.12	96.20	72.44	12.99	60.02	73.41
MHQ-PSL [24]	94.40	86.60	68.50	97.80	69.80	87.50	100.00	68.60	16.00	56.60	74.30
SGPLM-IR [25]	98.20	93.80	89.30	99.10	50.20	88.90	100.00	71.00	12.30	51.20	75.40
PIIFR [59]	98.63	94.70	76.45	83.92	61.39	82.47	100.00	78.16	19.55	57.24	75.20
HIMWR [47]	99.29	94.79	91.68	97.95	58.43	91.67	100.00	68.78	13.64	56.68	76.89
AE-IS [30]	98.30	94.20	72.40	100.00	56.80	83.60	98.40	76.80	18.20	62.40	76.10
IPS-YOLOm (Ours)	95.51	95.40	85.58	95.38	94.18	95.51	78.77	96.55	23.79	96.09	**85.68**

2) Online ICR: OICR [34] incorporates K ICR branches into WSDDN [39]. The previously obtained proposal feature vectors are fed into the k ICR branches to predict the CCS of proposals. The proposal with the highest CCS in the cth category and (k-1)th branch is regarded as the PGT of the cth category and kth ICR branch. So far, the loss function of the kth ICR branch, denoted as L_{ICR}^{k}, can be defined as follows:

$$L_{\mathrm{ICR}}^{k} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{c=1}^{C+1} w_n^k y_{c,n}^k \log x_{c,n}^k \tag{3}$$

where $(C+1)$th category indicates the background, $w_n^k = x_{\mathrm{cn}_c^{k-1}}^{k-1}$ represents the weight of r_n in the kth ICR branch, $x_{cn_c^{k-1}}^{k-1}$ refers to the CCS of $r_{n_c^{k-1}}$ in the cth category and $(k-1)$th ICR branch, n_c^{k-1} refers to the index of proposal with the highest CCS in the cth category and $(k-1)$th ICR branch, $x_{c,n}^k$ refers to the CCS of r_n in the cth category and kth ICR branch, $y_{c,n}^k = 1$ if $n = n_c^{k-1}$ or $r_n \in r_{n_c^{k-1}}^{\mathrm{nb}}$, and $r_{n_c^{k-1}}^{\mathrm{nb}}$ refers to the set of neighboring proposals of $r_{n_c^{k-1}}$, otherwise, $y_{c,n}^k = 0$.

3) Bounding Box Regression: To improve the localization accuracy of the WSOD model, K bounding box regression (BBR) branches are added to the OICR, and the loss function of the kth BBR branch, denoted as L_{BBR}^{k}, is as follows:

$$L_{\mathrm{BBR}}^{k} = -\frac{1}{Z^k}\sum_{z=1}^{Z^k}\left(\mathrm{smooth}_{L1}\left(b_r, \widehat{b_r}\right)\right), z \in \left[1, Z^k\right] \tag{4}$$

where Z^k refers to the number of positive samples in the kth BBR branch, $\mathrm{smooth}_{L1}(\cdot)$ denotes the smooth L1 function [40], and b_r and $\widehat{b_r}$ represent the predicted and actual offsets of the rth positive sample, respectively.

B. Other WSOD Methods

1) Traditional WSOD Models: The traditional MIL-based WSOD models can be categorized into two types in terms of the mining strategy of seed instances, where the seed instances and their neighboring instances are regarded as PGT instances which are used to train the ICR branches. The first kind of models rely solely on CCS to mine seed instances. For example, OICR [34] selects the proposals with the highest CCS as seed instances. MIST [41] chooses the proposals with top-K highest CCSs as seed instances instead of the proposals with the highest CCS. PCL [42] divides proposals into clusters based on their CCSs, and selects the cluster with the highest CCS as seed instances. Other related models include ITNet [43], CASD [44], and DCL [45].

The second kind of models integrate the CCS and other cues to mine seed instances. For instance, MHQ-PSL [24] combines the CCS with an object completeness score to obtain a proposal quality score of each proposal, and the seed instances are mined according to the proposal quality scores. SGPLM-IR [25] integrates the CCS with a class-specific overlap score to mine seed instances. OD-WSCL [46] constructs a feature

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TABLE V
COMPARISON WITH POPULAR METHODS ON THE DIOR DATASET IN TERMS OF MAP (%)

Method	Airplane	Airport	Baseball field	Basketball court	Bridge	Chimney	Dam	Expressway service area	Expressway toll station	Golf field	
WSDDN [39]	9.06	39.68	37.81	20.16	0.25	12.28	0.57	0.65	11.88	4.90	
OICR [34]	8.70	28.26	44.05	18.22	1.30	20.15	0.09	0.65	29.89	13.80	
PCL [42]	21.52	35.19	59.80	23.49	2.95	43.71	0.12	0.90	1.49	2.88	
MIST [41]	32.01	39.87	62.71	28.97	7.46	12.87	0.31	5.14	17.38	51.02	
DCL [45]	20.89	22.70	54.21	11.50	6.03	61.01	0.09	1.07	31.01	30.87	
PCIR [56]	30.37	36.06	54.22	26.60	9.09	58.59	0.22	9.65	36.18	32.59	
MIG [57]	22.20	52.57	62.76	25.78	8.47	67.42	0.66	8.85	28.71	57.28	
TCA [58]	25.13	30.84	62.92	40.00	4.13	67.78	8.07	23.80	29.89	22.34	
SAE [32]	20.57	62.41	62.65	23.54	7.59	64.62	0.22	34.52	30.62	55.38	
SPG [31]	31.32	36.66	62.79	29.10	6.08	62.66	0.31	15.00	30.10	35.00	
MHQ-PSL [24]	29.10	49.80	70.90	41.40	7.20	45.50	0.20	35.40	36.80	60.80	
SGPLM-IR [25]	39.10	64.60	64.40	26.90	6.30	62.30	0.90	12.20	26.30	55.30	
PIIFR [59]	32.90	70.50	63.20	45.70	0.20	69.70	0.20	12.40	39.40	56.40	
HIMWR [47]	42.19	65.01	66.15	25.74	6.70	60.15	1.29	13.48	25.31	57.81	
RINet [60]	26.20	57.40	62.70	25.10	9.90	69.20	1.40	13.30	36.20	51.40	
AE-IS [30]	31.80	50.90	63.20	29.40	8.90	68.70	1.30	15.10	35.50	51.60	
IPS-YOLOm (Ours)	55.40	50.24	71.20	30.50	1.01	68.20	1.22	16.05	37.30	52.68	
Method	Ground track field	Harbor	Overpass	Ship	Stadium	Storage tank	Tennis court	Train station	Vehicle	Windmill	mAP
WSDDN [39]	42.53	4.66	1.06	0.70	63.03	3.95	6.06	0.51	4.55	1.14	13.26
OICR [34]	57.39	10.66	11.06	9.09	59.29	7.10	0.68	0.14	9.09	0.41	16.50
PCL [42]	56.36	16.76	11.05	9.09	57.62	9.09	2.47	0.12	4.55	4.55	18.19
MIST [41]	49.48	5.36	12.24	29.43	35.53	25.36	0.81	4.59	22.22	0.80	22.18
DCL [45]	56.45	5.05	2.65	9.09	63.65	9.09	10.36	0.02	7.27	0.79	20.19
PCIR [56]	58.51	8.60	21.63	12.09	64.28	9.09	13.62	0.30	9.09	7.52	24.92
MIG [57]	47.73	23.77	0.77	6.42	54.13	13.15	4.12	14.76	0.23	2.43	25.11
TCA [58]	58.85	24.84	11.06	9.09	46.40	13.74	30.98	1.47	9.09	1.00	25.82
SAE [32]	52.70	17.57	6.85	9.09	51.59	15.43	1.69	14.44	1.41	9.16	27.10
SPG [31]	48.02	27.11	12.00	10.02	60.04	15.10	21.00	9.92	3.15	0.06	25.77
MHG-PSL [24]	48.50	14.00	25.10	18.50	48.90	11.70	11.90	3.50	11.30	1.70	28.60
SGPLM-IR [25]	60.60	9.40	23.10	13.40	57.40	17.70	1.50	14.00	11.50	3.50	28.50
PIIFR [59]	55.30	16.60	0.60	9.10	54.80	18.30	11.00	16.10	9.10	1.10	29.10
HIMWR [47]	61.42	10.41	20.40	14.14	58.62	18.91	2.16	13.61	10.02	4.69	28.91
RINet [60]	53.90	28.60	4.80	9.10	52.70	15.80	20.60	12.90	9.10	4.70	28.30
AE-IS [30]	52.30	28.80	13.30	11.20	56.90	16.30	22.40	14.00	8.00	2.60	29.10
IPS-YOLOm (Ours)	61.50	27.30	1.03	15.20	54.27	25.50	24.90	1.30	12.60	1.25	**30.40**

TABLE VI
COMPARISON WITH POPULAR METHODS ON THE DIOR DATASET IN TERMS OF CORLOC (%)

Method	Airplane	Airport field	Baseball field	Basketball court	Bridge	Chimney	Dam	Expressway service area	Expressway toll station	Golffield	
WSDDN [39]	5.72	59.88	94.24	55.94	4.92	23.40	1.03	6.79	44.52	12.75	
OICR [34]	15.98	51.45	94.77	55.79	3.55	23.89	0.00	4.82	56.68	22.42	
PCL [42]	61.14	46.86	95.39	63.61	7.32	95.07	0.21	5.71	5.14	50.77	
MIST [41]	91.60	53.20	93.50	66.30	10.80	30.70	1.50	14.30	35.20	47.50	
PCIR [56]	93.10	45.60	95.50	68.30	3.60	92.10	0.20	5.40	58.40	47.50	
MIG [57]	76.98	46.86	95.39	63.61	23.00	95.07	0.21	16.96	57.88	50.77	
TCA [58]	81.58	51.33	96.17	73.45	5.03	94.69	15.89	32.79	45.95	48.56	
SAE [32]	91.20	69.37	95.48	67.52	18.88	97.78	0.21	70.54	54.32	51.43	
SPG [31]	80.48	32.04	98.68	65.00	15.20	96.08	22.52	16.99	46.08	50.96	
MHQ-PSL [24]	85.50	68.90	96.80	75.80	11.60	94.70	0.80	67.50	60.50	46.50	
SGPLM-IR [25]	92.20	58.30	97.80	74.20	16.20	95.20	0.30	51.30	56.20	52.30	
PIIFR [59]	88.30	69.10	98.80	69.40	19.9	97.80	0.30	24.70	56.20	54.40	
HIMWR [47]	94.14	59.34	98.12	70.46	17.59	94.61	4.80	52.19	54.27	53.47	
RINet [60]	92.70	80.90	92.70	69.50	8.60	90.10	0.20	71.30	62.0	65.5	
AE-IS [30]	91.40	78.60	96.10	68.80	16.00	92.30	22.80	68.90	60.60	62.70	
IPS-YOLOm (Ours)	97.22	76.48	97.19	69.61	3.22	96.31	22.48	68.33	69.97	63.96	
Method	Ground track field	Harbor	Overpass	Ship	Stadium	Storage	Tennis court	Train station	Vehicle	Windmill	CorLoc
WSDDN [39]	89.90	5.45	10.00	22.96	98.54	79.61	15.06	3.45	11.56	3.22	32.44
OICR [34]	91.41	18.18	18.70	31.80	98.28	81.29	7.45	1.22	15.83	1.98	34.77
PCL [42]	89.39	42.12	19.78	37.94	97.93	80.65	13.77	0.20	10.50	6.94	41.52
MIST [41]	87.10	38.60	23.40	50.70	80.50	89.20	22.40	11.50	22.20	2.40	43.60
PCIR [56]	88.60	15.80	5.20	39.50	98.10	85.60	13.40	56.50	9.70	0.60	46.12
MIG [57]	89.39	42.12	19.78	37.94	97.93	80.65	13.77	10.34	10.50	6.94	46.80
TCA [58]	85.26	39.91	20.17	30.63	84.59	91.46	56.28	3.79	10.45	1.25	48.41
SAE [32]	88.28	48.03	2.28	33.56	14.11	83.35	65.59	19.88	16.41	2.85	49.42
SPG [31]	89.18	49.45	22.00	35.16	98.61	90.04	32.56	12.73	9.98	2.34	48.30
MHQ-PSL [24]	75.20	50.50	28.30	39.70	92.60	77.00	55.10	10.10	20.90	5.60	53.20
SGPLM-IR [25]	91.70	48.60	23.00	32.70	98.80	89.30	49.50	19.50	18.30	4.00	53.20
PIIFR [59]	89.60	49.10	19.40	34.50	96.70	84.70	63.20	15.60	11.60	3.40	52.30
HIMWR [47]	93.04	49.71	22.04	34.81	99.01	90.42	44.13	18.31	17.52	10.51	53.92
RINet [60]	85.10	51.40	15.70	44.60	98.60	80.30	14.80	22.70	6.90	2.60	52.80
AE-IS [30]	88.20	50.90	23.40	40.20	98.80	91.50	33.20	18.40	12.30	2.50	55.90
IPS-YOLOm (Ours)	90.77	50.67	15.59	42.08	97.21	90.32	34.33	14.52	20.08	3.53	**56.19**

bank for each category, and the feature bank is used to mine multiple seed instances for each category. Other related models include CICR [26], HIWMR [47], OPOD [48], and SDIM [49].

2) Pseudo-FSOD Models: The current pseudo-FSOD models can be classified into two categories, one is single-pass pseudo-FSOD model, another one is iterative pseudo-FSOD model. DCL [45] is a classical single-pass pseudo-FSOD model for WSOD in RSIs. First, DCL uses WSDDN to generate PGT, and the PGT is ranked from easy to hard through entropy-based difficulty evaluation. Afterward, to enhance model robustness, a Fast-RCNN model is progressively trained by using PGT ordered from easy to difficult. SGPLM-IR [25] is another advanced single-pass pseudo-FSOD model. First, the OICR model is improved by introducing the weakly supervised semantic segmentation technique, and the PGT is

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produced by the improved OICR. Then, the PGT is used to train a detection head in a fully supervised manner. Recently, WeakSAM [36] introduces an adaptive PGT generation scheme, and proposes an ROI drop regularization strategy to remove the PGT instances with large training loss during the process of training.

To our best knowledge, the existing iterative pseudo-FSOD model is SoS [35] used for natural scene images, and its procedure is summarized as follows. First, the PGT generated by the improved OICR is filtered by discarding the noisy PGT with large training loss, and the training samples can be divided into two parts, i.e., the labeled samples corresponding to clean PGT and the unlabeled samples corresponding to noisy PGT. Second, the labeled samples are used to train a teacher FSOD model, and the trained teacher FSOD model is used to annotate the unlabeled samples. Third, the labeled samples and the samples annotated by teacher FSOD model are jointly utilize to train a student FSOD model, where the samples annotated by teacher FSOD model are only used for training classification branch. Fourth, the parameters of teacher FSOD model are updated by the student FSOD model via exponential moving average algorithm [50]. Finally, the teacher FSOD model is iteratively trained by repeating the above step2–step4.

Note that, although the iterative pseudo-fully supervised training is conducted in the SoS model, the updating of labeled samples is not involved, which is the key difference between our IPS-YOLO and SoS. The single-pass pseudo-FSOD models are no exception. Furthermore, to our best knowledge, our IPS-YOLO is the first iterative pseudo-FSOD model for WSOD in RSIs.

III. Proposed Method

A. Overview

As illustrated in Fig. 2, the training of IPS-YOLO model can be divided into two stages. In the first stage, the baseline WSOD model is improved and trained in a weakly supervised manner, where the improvements focus on object proposals generation, the selection of augmented samples, and the definition of pseudo labels for ICR branches. In the second stage, a YOLO model is iteratively trained in a pseudo-fully supervised manner. Specifically, the detection results given by the improved baseline model are considered as initial PGT and is refined by the proposed FWSC strategy. Afterward, the refined PGT is employed to train a YOLO model, then the detection results given by the trained YOLO model are used to further update the PGT through the proposed FWSI strategy, and the above steps can be repeated multiple times to continuously enhance the quality of PGT. Therefore, the capability of trained YOLO model is enhanced accordingly.

B. Weakly Supervised Training of Improved Baseline WSOD Model

1) Proposal Generation Enhanced by SAM: The performance of SS is limited since it is a no-deep learning approach. Considering the significant impact of object proposals on final results, IPS-YOLO employs SAM to enhance the quality of proposals.

First, the input RSI is segmented by SAM to generate a series of segmented regions. Afterward, some rectangular bounding boxes, denoted as SAM-boxes, can be obtained, each of which tightly encloses a segmented region. Specifically, the top-left coordinates of each SAM-box are determined by the topmost and leftmost pixels of corresponding segmented region, and the bottom-right coordinates can be obtained in a similar way. Comparing with the object proposals generated by SS, the SAM-boxes can locate the objects more accurately due to the excellent performance of SAM in image segmentation. Therefore, both of them are aggregated to construct the final assembly of object proposals.

2) EPS of Augmented Training Samples: To enhance the robustness of baseline model, the training samples are augmented as follows: rotation (90°, 180°, 270°), horizontal and vertical flipping, and scaling (480, 576, 688, 864, 1000, 1200), which guarantees that the baseline model is invariant to rotation, flipping, and scaling. To ensure that each kind of augmented samples imposes same influence on the training process, four kinds of samples, i.e., original, rotated, flipped, and scaled samples, are randomly selected with equal probability. For example, the flipped samples are selected with a probability of 1/4 among four kinds of samples. Once the flipped samples are selected, horizontal or vertical flipped samples are selected with a probability of 1/2. In other words, the final probability that horizontal or vertical flipped samples are chosen is 1/8. The probability for other kind of samples can be calculated in a similar way.

3) Incorporating SLs Into the Definition of Pseudo Labels: So far, the one-hot label is usually used to define the pseudo label of each instance in ICR branches. The positive samples of each ICR branch consist of seed instances and their neighboring instances, where the seed instances are mined by OD-WSCL [46] in this article. Obviously, the localization accuracy of neighboring instances is less than the seed instances. Therefore, the neighboring instances should not be assigned the same labels as seed instances, and the novel definition of pseudo labels is as follows:

$$\mathrm{pl}_{c,n}^{k}=\begin{cases}1, & \text{if } r_n\in \mathrm{SI}_c^k\\ 1\times \mathrm{IoU}\left(r_n,\mathrm{si}_c^k\right), & \text{if } r_n\in NI_{\mathrm{si}_c^k}\\ 0, & \text{if } r_n\in \mathrm{BI}_c^k\end{cases} \tag{5}$$

where $\mathrm{pl}_{c,n}^{k}$ denotes the pseudo labels of r_n in the cth category and kth ICR branch, SI_c^k refers to the set of seed instances in the cth category and kth ICR branch, $\mathrm{si}_c^k \in \mathrm{SI}_c^k$ denotes the nearest seed instances of r_n, $\mathrm{NI}_{\mathrm{si}_c^k}$ represents the neighboring instances of si_c^k, and BI_c^k denotes the background instances in the cth category and kth ICR branch.

4) Training Loss of Improved Baseline WSOD Model: According to the novel definition of pseudo labels, the loss function of the kth ICR branch defined in (3) is rewritten as follows:

$$L_{\mathrm{ICR}}^{k}=-\frac{1}{N}\sum_{n=1}^{N}\sum_{c=1}^{C+1}w_n^k\mathrm{pl}_{c,n}^{k}\log \mathrm{x}_{c,n}^{k}. \tag{6}$$

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TABLE VII
COMPARISON WITH POPULAR METHODS ON THE NWPU VHR-10.V2 DATASET IN TERMS OF PARAMETERS, GFLOPS, AND MAP

Method	Parameters (M)	GFLOPS	mAP (%)
OICR [34]	134.60	286.23	34.52
MIST [41]	135.02	290.43	51.52
PISLM [24]	135.53	296.43	63.80
SGPLM-IR [25]	135.63	334.26	65.20
RINet [60]	135.11	534.26	70.40
Ours:			
IPS-YOLOn	0.30	8.10	72.70
IPS-YOLOs	11.00	28.50	73.70
IPS-YOLOm	25.00	78.70	76.20
IPS-YOLOl	43.00	164.90	76.80

The final training loss of improved baseline WSOD model, denoted as L_B, is as follows:

$$L_B = L_W + \frac{1}{K}\sum_{k=1}^{K}\left(L_{\text{ICR}}^{k} + L_{\text{BBR}}^{k}\right). \tag{7}$$

C. Iterative Pseudo-Fully Supervised Training of YOLO Model

The main content of this section includes the generation and IR of PGT. With the iterative updating of PGT, the YOLO model is iteratively trained in a pseudo-fully supervised way.

1) PGT Generation Based on FWSC Strategy: After completing the weakly supervised training of improved baseline model, the training samples are imported into the trained baseline model to infer their detection results, where the final CCS of each proposal is the average value of K CCSs output from K ICR branches, and the final offsets of each proposal are obtained in the same way, as shown in Fig. 2. The detection results given by the improved baseline model are considered as initial PGT. Subsequently, the proposed FWSC strategy is used to refine the initial PGT, and the refined PGT is utilized to train the YOLO model.

As shown in Fig. 2, the FWSC strategy includes filtering and WS-CCS operations. The filtering operation is used to remove the instances with low CCS. Part of the role of WS-CCS operation is similar to that of NMS [51] operation, i.e., removing the instances with high spatial overlap. However, there is an intrinsic difference between WS-CCS and NMS, i.e., any two instances which have the same category and high spatial overlap are weighted and synthesized according to their CCSs instead of eliminating the one with lower CCS, and the motivation of this operation is as follows.

As mentioned before, the PGT generated by traditional WSOD models typically focus on the most salient regions of target rather than the whole target. Considering the fact that any two proposals with same category and high spatial overlap usually belong to the same target, the two proposals are fused through WS-CCS operation to cover the whole target as much as possible.

The details of filtering and WS-CCS operations are shown in Algorithms 1 and 2, respectively. First, the initial PGT instances and their CCSs, denoted as P and S, are imported into Algorithm 1 to obtain the filtered results and their CCSs, denoted as PF and SF. Afterward, the PF and SF are imported into Algorithm 2 to get the refined PGT and corresponding CCSs, denoted as PR and SR.

Algorithm 1 Filtering Operation

Input: $P = \{P_1, \ldots, P_c, \ldots, P_C\}$ // P_c denotes the set of initial PGT instances in cth category
$S = \{S_1, \ldots, S_c, \ldots, S_C\}$ // S_c denotes the CCSs of P_c
T_{CCS} // CCS threshold
Output: $PF = \{PF_1, \ldots, PF_c, \ldots, PF_C\}$ // PF_c denotes the filtered results of P_c
$SF = \{SF_1, \ldots, SF_c, \ldots, SF_C\}$ // SF_c denotes the CCSs of PF_c

$PF = SF = \emptyset$
for $c = 1$ to C **do**
  $idm = \arg\max_i(S_{c,i})$ // $S_{c,i}$ denotes the CCS of $P_{c,i}$, $P_{c,i}$ denotes the ith instance in P_c
  if $S_{c,idm} < T_{CCS}$ **then**
    $PF_c = PF_c \cup P_{c,idm}$
  end if
  if $S_{c,idm} \geq T_{CCS}$ **then**
    for $i = 1$ to $|P_c|$ **do**
      if $S_{c,i} \geq T_{CCS}$ **then**
        $PF_c = PF_c \cup P_{c,i}$, $SF_c = SF_c \cup S_{c,i}$
      end if
    end for
  end if
end for

2) IR of PGT Based on FWSI Strategy: To further improve the quality of PGT, the proposed FWSI strategy is used to iteratively refine the PGT in this section. First, the PR obtained in Algorithm 2 is employed to train the YOLO model. Next, the training samples are imported into the trained YOLO model to infer the detection results and CCS, denoted as PY and SY. Finally, the PR, SR, PY, and SY are imported into the FWSI strategy to refine the PGT again, obtaining iteratively refined PGT and their CCSs, denoted as PIR and SIR. The above steps can be repeated multiple times, i.e., PR = PIR and SR = SIR.

As shown in Fig. 2, the FWSI strategy consists of filtering and WS-IoU operations, where the filtering operation has been introduced in Algorithm 1. The WS-IoU operation uses the detection results produced by the YOLO model to update the PGT, where the results which have very high spatial overlap with PGT instances are weighted and fused with their nearest PGT instances according to the IoU between them; on the contrary, the results are directly added to the set of PGT instances, otherwise, they are ignored. The FWSI strategy employs IoU instead of CCSs (SR and SY) for weighted synthesis, and the primary reason is as follows. The SR and SY are generated from the improved baseline model and YOLO model, respectively. In other words, the SR and SY cannot be compared with each other in the same level.

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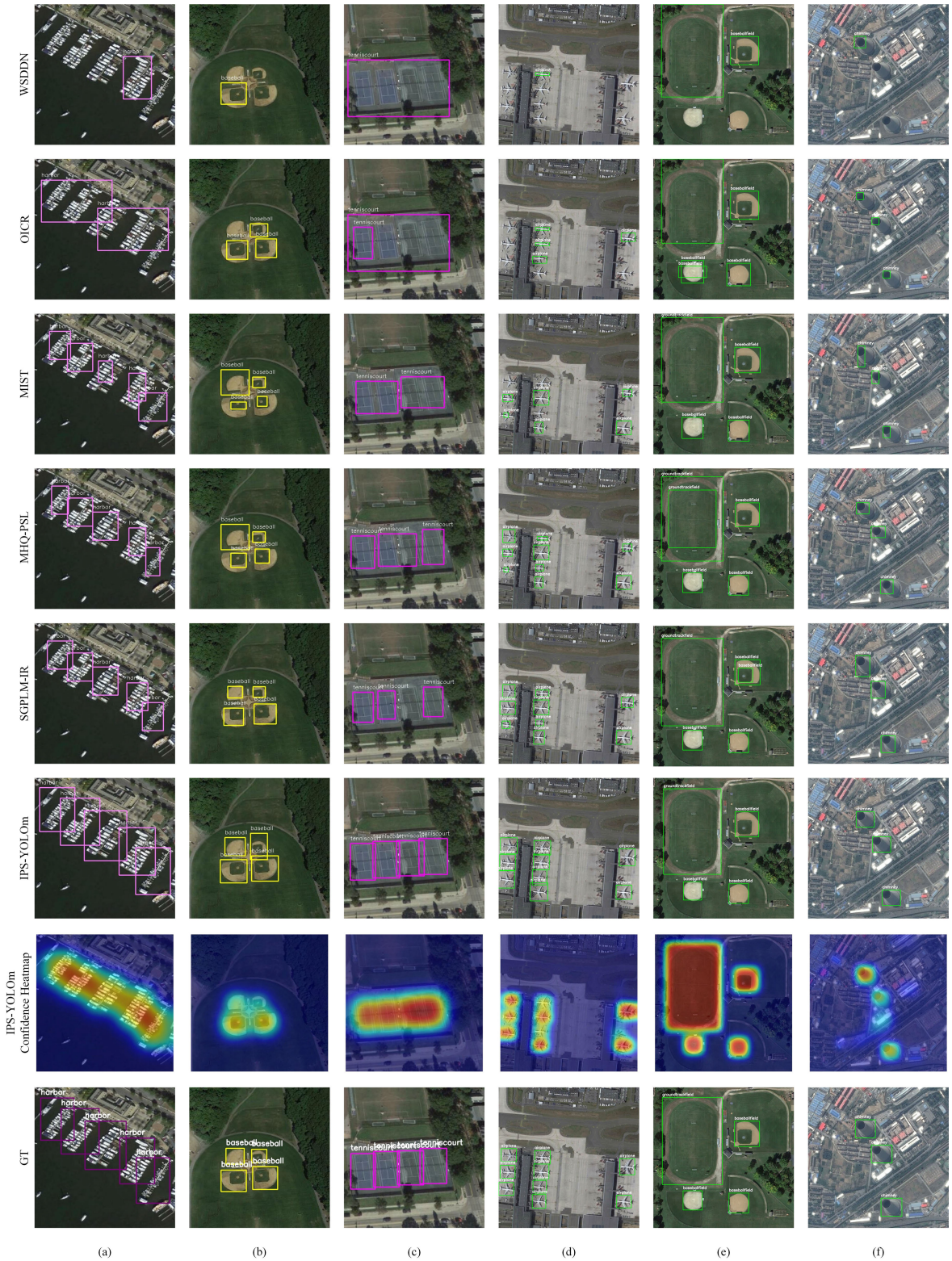


Fig. 4. Subjective comparison with other WSOD models, along with confidence heatmaps of our IPS-YOLOm. (a)–(f) Selected from NWPU VHR-10.v2 and DIOR datasets, respectively.

The details of WS-IoU operation are shown in Algorithm 3. Specifically, the PY and SY are first imported into Algorithm 1 to obtain the filtered results and their CCSs, denoted as PFY and SFY, and then the PR, SR, PFY, and SFY are imported into Algorithm 3 to obtain the PIR and SIR.

3) Training and Inference of YOLO: As mentioned in Section III-C2, the YOLO model is iteratively trained along with the IR of PGT. The original loss function of the YOLO model is also adopted in this article. Any RSI is fed into the trained YOLO model for inference when the iterative training is finished.

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(a) (b) (c) (d)

Fig. 5. Illustration of some failure detections. (a) Selected from NWPU VHR-10.v2 dataset, (b), (c), and (d) Selected from DIOR dataset.

IV. Experiments

A. Datasets and Evaluation Metrics

Experiments are conducted on the NWPU VHR-10.v2 [52] and DIOR datasets [53]. The NWPU VHR-10.v2 dataset consists of 2775 instances and 1172 RSIs with a resolution of 400 × 400, covering ten categories: airplane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge, and vehicle. The quantities of training, validation, and testing samples are 679, 200, and 293, respectively. The training and validation subsets are used for training. The DIOR dataset contains 192 472 instances and 23 463 RSIs with a resolution of 800 × 800, covering 20 categories: airplane, airport, baseball diamond, basketball court, bridge, chimney, dam, expressway toll station, expressway service area, golf course, ground track field, harbor, overpass, ship, stadium, storage tank, tennis court, train station, vehicle, and windmill. The quantities of trainval and testing samples are 11 725 and 11 738, respectively. The trainval subset is used for training. The mean average precision (mAP) is used to evaluate the detection accuracy on the testing subset, while correct localization (CorLoc) is employed to measure the localization performance on the training and validation subsets. The GFOLOPS and number of learnable parameters (denoted as parameters in Section IV-G) are utilized to evaluate computational cost.

B. Implementation Details

Our model adopts YOLOv8m [54] as baseline model, denoted as IPS-YOLOm. The VGG16 [55] pretrained on ImageNet is used as the backbone of baseline model. The number of ICR and regression branches is set to 3, i.e., $K = 3$. The total number of iterations of baseline model is set to 30 K (60 K) on the NWPU VHR-10.v2 (DIOR) dataset, and the initial learning rate is set to 0.1, which is reduced to 10% of its previous value at 20 K (50 K) and 26 K (56 K) iterations on the NWPU (DIOR) dataset, respectively. The momentum, weight decay, and batch size are set to 0.9, 0.05, and 8, respectively. The training setting and NMS threshold are consistent with that one of YOLOv8m model. Our model is trained on Ubuntu 22.04 with 8 GPUs (Titan RTX GPUs, 24GB memory).

C. Parameter Analysis

The T_{CCS} defined in Algorithm 1, T_L, T_H defined in Algorithm 2, and the number of IRs of PGT, denoted as NIR, are quantitatively analyzed in this section. As shown in Fig. 3, the mAP of IPS-YOLO achieves the highest value on both NWPU VHR-10.v2 and DIOR datasets when T_{CCS}, T_L, T_H, and NIR are set to 0.3, 0.4, 0.8, and 2, respectively. Note that NIR = 0 denotes that our model is trained by single-pass refined PGT.

D. Ablation Study

1) Ablation Study of Improved Baseline Model: The ablation study of improved baseline model is carried out to evaluate the effectiveness of SAM-guided proposal generation strategy (SAMPG) mentioned in Section III-B1, equal probability selection (EPS) strategy mentioned in Section III-B2, and SL mentioned in Section III-B3. As shown in Table I, the mAP of baseline model can be increased by 8.16% (3.64%), 8.78% (4.1%), and 10.37% (4.32%) by using SAMPG, EPS, and SL alone, which validates the effectiveness of above three strategies. The mAP is increased by 13.67% (5.02%) when the three strategies are all adopted on the NWPU VHR-10.v2 (DIOR) dataset, demonstrating that the combination of the above three strategies is effective.

2) Ablation Study of FWSC and IR: Another ablation study is conducted to evaluate the effectiveness of FWSC strategy mentioned in Section III-C1, and IR strategy mentioned in

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Algorithm 2 WS-CCS Operation

Input: $PF = \{PF_1, \ldots, PF_c, \ldots, PF_C\}$
$SF = \{SF_1, \ldots, SF_c, \ldots, SF_C\}$
T_L, T_H // T_L and T_H denote the low and high IoU thresholds, respectively
Output: $\mathrm{PR} = \{\mathrm{PR}_1, \ldots, \mathrm{PR}_c, \ldots, \mathrm{PR}_C\}$ // PR_c denotes the refined results of PF_c
$\mathrm{SR} = \{\mathrm{SR}_1, \ldots, \mathrm{SR}_c, \ldots, \mathrm{SR}_C\}$ // SR_c denotes the CCSs of PR_c

$\mathrm{PR} = \mathrm{SR} = \emptyset$
for $c = 1$ to C **do**
  while $PF_c \neq \emptyset$ **do**
    $IB = \emptyset$ // IB denotes the index buffer
    $idm = \arg\max_i(SF_{c,i})$ // $SF_{c,i}$ denotes the CCS of $PF_{c,i}$, $PF_{c,i}$ denotes the ith instance in PF_c
    for $i = 1$ to $|PF_c|$ **do**
      if $IoU(PF_{c,idm}, PF_{c,i}) \leq T_L$ **then**
        $\mathrm{PR}_c = \mathrm{PR}_c \cup PF_{c,i}$, $\mathrm{SR}_c = \mathrm{SR}_c \cup SF_{c,i}$
      end if
      if $IoU(PF_{c,idm}, PF_{c,i}) \geq T_H$ **then**
        $IB = IB \cup i$
      end if
      remove $PF_{c,i}$, $SF_{c,i}$ from PF_c, SF_c
    end for
    $M = |IB|$ // $|IB|$ denotes the cardinality of IB
    $w_j = \frac{SF_{c,IB\{j\}}}{\sum_{j=1}^{M} SF_{c,IB\{j\}}}$// w_j denotes the weight of $PF_{c,IB\{j\}}$
    $Cor = \sum_{j=1}^{M} w_j Cor_PF_{c,IB\{j\}}$
    // $Cor_PF_{c,IB\{j\}}$ denotes the coordinate of $PF_{c,IB\{j\}}$, Cor denotes the coordinates of instances produced by the weighted synthesis
    $\mathrm{PR}_c = \mathrm{PR}_c \cup PWS_c^{cor}$
    // PWS_c^{cor} denotes the instance with Cor in cth category
    $SWS_c^{cor} = \sum_{j=1}^{M} w_j SF_{c,IB\{j\}}$
    // SWS_c^{cor} denotes the CCS of PWS_c^{cor}
    $\mathrm{SR}_c = \mathrm{SR}_c \cup SWS_c^{cor}$
  end while
end for

Algorithm 3 WS-IoU Operation

Input: $\mathrm{PR} = \{\mathrm{PR}_1, \ldots, \mathrm{PR}_c, \ldots, \mathrm{PR}_C\}$
$\mathrm{SR} = \{\mathrm{SR}_1, \ldots, \mathrm{SR}_c, \ldots, \mathrm{SR}_C\}$
$PFY = \{PFY_1, \ldots, PFY_c, \ldots, PFY_C\}$ // PFY_c denotes the subset of PFY in cth category
$SFY = \{SFY_1, \ldots, SFY_c, \ldots, SFY_C\}$ // SFY_c denotes the CCSs of PFY_c
T_L, T_H
Output: $PIR = \{PIR_1, \ldots, PIR_c, \ldots, PIR_C\}$ // PIR_c denotes the refined results of PR_c
$SIR = \{SIR_1, \ldots, SIR_c, \ldots, SIR_C\}$ // SIR_c denotes the CCSs of PIR_c

$PIR = SIR = \emptyset$
for $c = 1$ to C **do**
  for $i = 1$ to $|PFY_c|$ **do**
    $idm = \arg\max_j IoU(PFY_{c,i}, \mathrm{PR}_{c,j})$
    // $PFY_{c,i}$ ($\mathrm{PR}_{c,j}$) denotes the ith (jth) instance in PFY_c (PR_c)
    $MIoU = IoU(PFY_{c,i}, \mathrm{PR}_{c,idm})$
    if $MIoU \leq T_L$ **then**
      $PIR_c = PIR_c \cup PFY_{c,i}$, $SIR_c = SIR_c \cup SFY_{c,i}$
      // $SFY_{c,i}$ denotes the CCS of $PFY_{c,i}$
    end if
    if $MIoU \geq T_H$ **then**
      $CWS = \frac{1}{1+MIoU} Cor_\mathrm{PR}_{c,idm} + \frac{MIoU}{1+MIoU} Cor_PFY_{c,i}$
      // $Cor_\mathrm{PR}_{c,idm}$ ($Cor_PFY_{c,i}$) denotes the coordinates of $\mathrm{PR}_{c,idm}$ ($PFY_{c,i}$), CWS denotes the coordinates of instance produced by the weighted synthesis
      $PIR_c = PIR_c \cup PW_c^{CWS}$ // PW_c^{CWS} denotes the instance with CWS in cth category
      $SW_c^{CWS} = \frac{1}{1+MIoU} \mathrm{SR}_{c,idm} + \frac{MIoU}{1+MIoU} SFY_{c,i}$
      // SW_c^{CWS} denotes the CCS of PW_c^{CWS}
      $SIR_c = SIR_c \cup SW_c^{CWS}$
      remove $\mathrm{PR}_{c,idm}$, $\mathrm{SR}_{c,idm}$ from PR_c, SR_c
      // $\mathrm{PR}_{c,idm}$ denotes the CCS of $\mathrm{PR}_{c,idm}$
    end if
    remove $PFY_{c,i}$, $SFY_{c,i}$ from PFY_c, SFY_c
  end for
  $PIR_c = PIR_c \cup \mathrm{PR}_c$
  $SIR_c = SIR_c \cup \mathrm{SR}_c$
end for

Section III-C2. As shown in Table II, the mAP of improved baseline model is increased by 7.11% (2.18%) and 8.41% (2.3%) by using the FWSC and IR strategies alone, and the mAP is increased by 11.01% (3.2%) when the two strategies are all adopted on the NWPU VHR-10.v2 (DIOR) dataset, demonstrating that the two strategies and their combination are effective.

E. Comparison With Popular Methods on the NWPU VHR-10.v2 Dataset

As shown in Tables III and IV, compared with 16 WSOD methods, including WSDDN [39], OICR [34], PCL [42], MIST [41], DCL [45], PCIR [56], MIG [57], TCA [58], SAE [32], SPG [31], MHQ-PSL [24], SGPLM-IR [25], PIIFR [59], HIMMR [47], RINet [60], and AE-IS [30], the mAP (CorLoc) of IPS-YOLO is improved by 41.08% (50.44%), 41.68% (45.67%), 36.79% (40.62%), 24.68% (15.38%), 24.09%, 21.23% (13.81%), 20.25% (15.52%), 17.38% (12.92%), 15.48% (12.22%), 13.4% (12.27%), 12.4% (11.38%), 11% (10.28%), 11% (10.48%), 9.96% (8.79%),

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5.8%, and 4.2% (9.58%), separately. Obviously, the above comparisons demonstrate the SOTA performance of our IPS-YOLO model on the NWPU VHR-10.v2 dataset.

F. Comparison With Popular Methods on the DIOR Dataset

To further evaluate the overall effectiveness of IPS-YOLO, the comparison with 16 WSOD methods mentioned in Section IV-E is conducted on the more challenging DIOR dataset. As shown in Tables V and VI, compared with WSDDN, OICR, PCL, MIST, DCL, PCIR, MIG, TCA, SAE, SPG, MHQ-PSL, SGPLM-IR, PIIFR, HIMMR, RINet, and AE-IS, the mAP (CorLoc) of IPS-YOLO is improved by 17.14% (23.75%), 13.9% (21.42%), 12.21% (14.67%), 8.22% (12.59%), 10.21%, 5.48% (10.07%), 5.29% (9.39%), 4.58% (7.78%), 3.3% (6.77%), 4.63% (7.89%), 1.8% (2.99%), 1.9% (2.99%), 1.3% (3.89%), 1.49% (2.27%), 2.1% (3.39%), and 1.3% (0.29%). Obviously, the above comparisons demonstrate the superiority of our IPS-YOLO model on the DIOR dataset again.

G. Comparison With Popular Methods in Terms of Computational Cost

To evaluate the impact of different baseline models on our IPS-YOLO model, the comprehensive quantitative comparisons are conducted between four variants of IPS-YOLO, i.e., IPS-YOLOn, IPS-YOLOs, IPS-YOLOm, and IPS-YOLOl, which are based on YOLOv8n, YOLOv8s, YOLOv8m, and YOLOv8l, respectively. As shown in Table VII, IPS-YOLOv8m achieves relatively lower computational costs while maintaining high mAP value, demonstrating optimal comprehensive performance. Consequently, the IPS-YOLOv8m is selected as our final pseudo-FSOD model.

To further evaluate the computational cost of the proposed IPS-YOLO, the parameters and GFLOPS are compared with those of popular methods, as shown in Table VII. The four variants of IPS-YOLO, e.g., IPS-YOLOn, IPS-YOLOs, IPS-YOLOm, and IPS-YOLOl achieve the lower parameters and GFLOPS while obtaining the higher mAP values, which clearly demonstrates the superior performance of IPS-YOLO in terms of both computational efficiency and detection accuracy.

H. Subjective Evaluation

To intuitively evaluate the effectiveness of our IPS-YOLOm, this section conducts subjective comparisons between IPS-YOLOm and other WSOD models and presents the confidence heatmap of IPS-YOLOm. As shown in Fig. 4(a), the other WSOD models usually focus on the most salient part of harbors, while our IPS-YOLOm can localize harbors completely. The similar occasions can also be seen in Fig. 4(b), (e), and (f), which validate the effectiveness of our FWSC strategy. As shown in Fig. 4(d), some densely distributed airplanes are omitted by other WSOD models while our IPS-YOLOm can detect all of airplanes. The similar occasions can also be seen in Fig. 4(c), which validates the effectiveness of our IR strategy. Additionally, the confidence heatmaps of IPS-YOLOm can accurately focus on the detected targets, validating the overall effectiveness of IPS-YOLOm.

I. Analysis of Failure Detection Results

Although the IPS-YOLO model demonstrates competitive performance compared to other WSOD models, it still faces challenges in certain scenarios. Some failure detections and their possible reasons are as follows.

1) Challenges From Symbiotic Context: As shown in Fig. 5(a) and (b), IPS-YOLO mistakenly detects both the target and its neighboring context as the target, which is a hard problem for WSOD models driven by image-level labels because only the targets and their strong contextual neighboring regions can jointly determine the image-level labels in certain scenarios.

2) Challenges From Shadow: As illustrated in Fig. 5(c) and (d), IPS-YOLO incorrectly identifies the shadow of target as the part of target or the target itself, which is induced by the similar shape between the targets and their shadows.

V. Conclusion

A novel WSOD model named IPS-YOLO is introduced in this article, and its major contribution includes three aspects. First, an FWSC strategy is proposed to generate PGT for addressing the issue that current pseudo-FSOD models tend to focus on the most salient parts rather than the whole object. Specifically, the FWSC strategy filters initial PGT instances according to their CCSs, and then imposes a WS-CCS operation on the filtered PGT instances, where any two instances whose spatial overlap is very high, are weighted and synthesized according to their CCSs for capturing the whole object as much as possible. Second, an IR scheme of PGT is proposed to handle the issue that the quality of PGT cannot be continuously improved along with the training of pseudo-FSOD model. Specifically, the PGT produced by the FWSC strategy is first used to train a YOLO model, then the detection results inferred from the trained YOLO model are employed to refine the PGT by using the proposed FWSI strategy, and the above steps can be repeated multiple times to continuously enhance the quality of PGT. Third, the proposed SAMPG, EPS, and SL strategies are used to improve the traditional baseline WSOD model in proposals generation, the selection of augmented samples, and the definition of pseudo labels, where the improved baseline model is used to generate initial PGT instances. Ablation studies validate the effectiveness of SAMPG, EPS, SL, FWSC, IR, and their combination. The comparisons with popular WSOD methods show that the 76.2% (30.4%) mAP and 85.68% (56.19%) CorLoc of IPS-YOLO are superior to those of popular methods on the NWPU VHR-10.v2 (DIOR) dataset.

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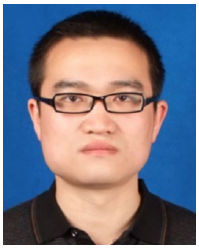
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